

14. Two number are in the ratio 3 : 5. If 9 is subtracted from each, the new numbers are in the ratio 12 : 23. The smaller number is:

Ⓐ 27

Ⓑ 33 ✓

Ⓒ 49

Ⓓ 55

Answer: Option Ⓑ

Explanation:

Let the numbers be $3x$ and $5x$.

$$\text{Then, } \frac{3x - 9}{5x - 9} = \frac{12}{23}$$

$$\Rightarrow 23(3x - 9) = 12(5x - 9)$$

$$\Rightarrow 9x = 99$$

$$\Rightarrow x = 11.$$

∴ The smaller number = $(3 \times 11) = 33$.

21. How many seconds will a 500 metre long train take to cross a man walking with a speed of 3 km/hr in the direction of the moving train if the speed of the train is 63 km/hr?

Ⓐ 25

Ⓑ 30 ✓

Ⓒ 40

Ⓓ 45

Answer: Option Ⓑ

Explanation:

of the train relative to man = $(63 - 3)$ km/hr

= 60 km/hr

$$= \left(60 \times \frac{5}{18} \right) \text{ m/sec}$$

$$= \left(\frac{50}{3} \right) \text{ m/sec.}$$

$$\text{e taken to pass the man} = \left(500 \times \frac{3}{50} \right) \text{ sec}$$

= 30 sec.

Two dice are tossed. The probability that the total score is a prime number is:

A $\frac{1}{12}$

Correct Answer

B $\frac{5}{12}$

C $\frac{7}{12}$

D $\frac{11}{12}$

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Solution



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Clearly, $n(S) = (6 \times 6) = 36$.

Let E = Event that the sum is a prime number.

Then $E = \{ (1, 1), (1, 2), (1, 4), (1, 6), (2, 1), (2, 3), (2, 5), (3, 2), (3, 4), (4, 1), (4, 3), (5, 2), (5, 6), (6, 1), (6, 5) \}$

$n(E) = 15$.

Hence, the probability that the total score is a prime number is

$$P(E) = \frac{n(E)}{n(S)} = \frac{15}{36} = \frac{5}{12}$$

15. Three pipes A, B and C can fill a tank in 6 hours. After working at it together for 2 hours, C is closed and A and B can fill the remaining part in 7 hours. The number of hours taken by C alone to fill the tank is:

Ⓐ 10

Ⓑ 12

Ⓒ 14 ✓

Ⓓ 16

Answer: Option Ⓒ

Explanation:

$$\text{Part filled in 2 hours} = \frac{2}{6} = \frac{1}{3}$$

$$\text{Remaining part} = \left(1 - \frac{1}{3} \right) = \frac{2}{3}.$$

$$\therefore (A + B)\text{'s 7 hour's work} = \frac{2}{3}$$

$$(A + B)\text{'s 1 hour's work} = \frac{2}{21}$$

$$\therefore \text{C's 1 hour's work} = \{ (A + B + C)\text{'s 1 hour's work} \} - \{ (A + B)\text{'s 1 hour's work} \}$$

$$= \left(\frac{1}{6} - \frac{2}{21} \right) = \frac{1}{14}$$

$\therefore$ C alone can fill the tank in 14 hours.

17. How many times do the hands of a clock coincide in a day?

Ⓐ 20

Ⓑ 21

Ⓒ 22 ✓

Ⓓ 24

Answer: Option Ⓒ

Explanation:

The hands of a clock coincide 11 times in every 12 hours (Since between 11 and 1, they coincide only once, *i.e.*, at 12 o'clock).

AM

12:00

1:05

2:11

3:16

4:22

5:27

6:33

7:38

8:44

9:49

10:55

PM

12:00

1:05

2:11

3:16

4:22

5:27

6:33

7:38

8:44

9:49

10:55

The hands overlap about every 65 minutes, not every 60 minutes.

∴ The hands coincide 22 times in a day.

5. Excluding stoppages, the speed of a bus is 54 kmph and including stoppages, it is 45 kmph. For how many minutes does the bus stop per hour?

- Ⓐ 9
- Ⓑ 10 
- Ⓒ 12
- Ⓓ 20

Answer: Option Ⓑ

Explanation:

to stoppages, it covers 9 km less.

$$\therefore \text{taken to cover 9 km} = \left(\frac{9}{54} \times 60 \right) \text{min} = 10 \text{ min.}$$

7. A man complete a journey in 10 hours. He travels first half of the journey at the rate of 21 km/hr and second half at the rate of 24 km/hr. Find the total journey in km.

- Ⓐ 220 km
- Ⓑ 224 km ✓
- Ⓒ 230 km
- Ⓓ 234 km

Answer: Option Ⓑ

Explanation:

$$\frac{(1/2)x}{21} + \frac{(1/2)x}{24} = 10$$

$$\Rightarrow \frac{x}{21} + \frac{x}{24} = 20$$

$$\Rightarrow 15x = 168 \times 20$$

$$\Rightarrow x = \left(\frac{168 \times 20}{15} \right) = 224 \text{ km.}$$

4 mat-weavers can weave 4 mats in 4 days. At the same rate, how many mats would be woven by 8 mat-weavers in 8 days?

A 4

B 8

C 12

D 16

Correct Answer

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Solution

 Verified by Toppr

Let the required number of mats be x .

Clearly, if number of weavers are increase then number of mats also increase.

So, the given situation is in Direct Proportion.

Similarly, More days, More mats (Direct Proportion)

$$\begin{array}{l} \text{Weavers} \quad 4 : 8 \\ \text{Days} \quad \quad 4 : 8 \end{array} \} :: 4 : x$$

$$\therefore 4 \times 4 \times x = 8 \times 8 \times 4$$

$$\Rightarrow x = \frac{(8 \times 8 \times 4)}{(4 \times 4)}$$

$$\Rightarrow x = 16$$

Hence, Option B is correct.

14, 28, 20, 40, 32, 64, ?

A 52

B 56

C 128

D 48

The correct Answer is:

B

$$14 \times 2 = 28$$

$$28 - 8 = 20$$

$$20 \times 2 = 40$$

$$40 - 8 = 32$$

$$32 \times 2 = 64$$

$$64 - 8 = 56$$

Q: The total age of A and B is 12 years more than the total age of B and C. C is how many years younger than A ?

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A) 12

B) 13

C) 14

D) 15

Answer: A) 12

Explanation:

$$(A+B) - (B+C) = 12$$

$$\Rightarrow A - C = 12.$$

$\Rightarrow$ C is younger than A by 12 years.

Q1. What is the output for the provided input in the program below?

```
1 Input: s = "abca"
2
3 class Solution {
4     public boolean zoho(String s) {
5         int i = 0;
6         int j = s.length() - 1;
7         while(i <= j){
8             if(s.charAt(i) == s.charAt(j)){
9                 i++;
10                j--;
11            }
12            else return help(s, i + 1, j) || help(s, i, j - 1);
13        }
14        return true;
15    }
16    public boolean help(String s, int i, int j){
17        while(i <= j){
18            if(s.charAt(i) == s.charAt(j)){
19                i++;
20                j--;
21            }
22            else return false;
23        }
24        return true;
25    }
26 }
```



* 6) What is the output for the provided input in the program below?

```
1 Input: n = 5
2
3 public int[] zoho(int num) {
4     int[] f = new int[num + 1];
5     for (int i=1; i<=num; i++) f[i] = f[i >> 1] + (i & 1);
6     return f;
7 }
```

* 7) What is the output for the provided input in the program below?

7) What is the output for the provided input in the program below?

```
1 Input: arr = [3,1,7,11]
2
3 class Solution {
4     public boolean zoho(int[] arr) {
5         Arrays.sort(arr);
6         for (int i = 0; i < arr.length; i++) {
7             int target = 2 * arr[i];
8             int lo = 0, hi = arr.length - 1;
9             while (lo <= hi) {
10                 int mid = lo + (hi - lo) / 2;
11                 if (arr[mid] == target && mid != i)
12                     return true;
13                 if (arr[mid] < target)
14                     lo = mid + 1;
15                 else
16                     hi = mid - 1;
17             }
18         }
19         return false;
20     }
21 }
```


/zohocorp/bcB3zW

YouTube

Q) What is the output for the provided input in the program below?

```

1 Input: nums = [2,2,3,1]
2
3 class Solution {
4     public int zoho(int[] nums) {
5         Pair<Integer, Boolean> a = new Pair<Integer, Boolean>(-1,
6         false);
7         Pair<Integer, Boolean> b = new Pair<Integer, Boolean>(-1,
8         false);
9         Pair<Integer, Boolean> c = new Pair<Integer, Boolean>(-1,
10        false);
11        for (int num : nums) {
12            if ((a.getValue() && a.getKey() == num) ||
13                (b.getValue() && b.getKey() == num) ||
14                (c.getValue() && c.getKey() == num)) {
15                continue;
16            }
17            if (!a.getValue() || a.getKey() <= num) {
18                c = b;
19                b = a;
20                a = new Pair<Integer, Boolean>(num, true);
21            }
22            else if (!b.getValue() || b.getKey() <= num) {
23                c = b;
24                b = new Pair<Integer, Boolean>(num, true);
25            }
26            else if (!c.getValue() || c.getKey() <= num)
27                c = new Pair<Integer, Boolean>(num, true);
28        }
29        if (!c.getValue())
30            return a.getKey();
31        return c.getKey();
32    }
33 }

```

3, 7

* 8) What is the output for the provided input in the program :

```
1 Input: nums = [0,2,1,5,3,4]
2
3 public int[] buildArray(int[] nums) {
4     int[] ans = new int[nums.length];
5     for(int i = 0; i < nums.length; i++)
6         ans[i] = nums[nums[i]];
7     return ans;
8 }
```


0, 1, 2, 4, 5, 3

* 9) What is the output for the provided input in the program below?

```
1 Input: str1 = "aA", str2 = "aAAbbbb"
2
3 class Solution {
4     public int zoho(String str1, String str2) {
5         int num = 0;
6         for (int i = 0 ; i < str2.length(); i ++) {
7             if(str1.indexOf(str2.charAt(i)) != -1)
8                 num++;
9         }
10        return num;
11    }
12 }
```

5

What is the output for the provided input in the program below?

```
1 Input: arr = [1,0,0,0,1], n = 1
2
3 public class Solution {
4     public boolean zoho(int[] arr, int n) {
5         int count = 0;
6         for (int i = 0; i < arr.length; i++) {
7             if (arr[i] == 0) {
8                 boolean l = (i == 0) || (arr[i - 1] == 0);
9                 boolean r = (i == arr.length - 1) || (arr[i + 1] ==
10                    0);
11                 if (l && r) {
12                     arr[i] = 1;
13                     count++;
14                 }
15             }
16             return count >= n;
17         }
18     }
```

* 4) What is the output for the provided input in the program below?

```
1 Input: nums = [1,3,5,6], target = 7
2
3 public int zoho(int[] A, int target) {
4     int low = 0, high = A.length-1;
5     while(low<=high){
6         int mid = (low+high)/2;
7         if(A[mid] == target) return mid;
8         else if(A[mid] > target) high = mid-1;
9         else low = mid+1;
```